

Eddu Suchier - 192524070

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

DSA06 – DATA HANDLING AND VISUALIZATION

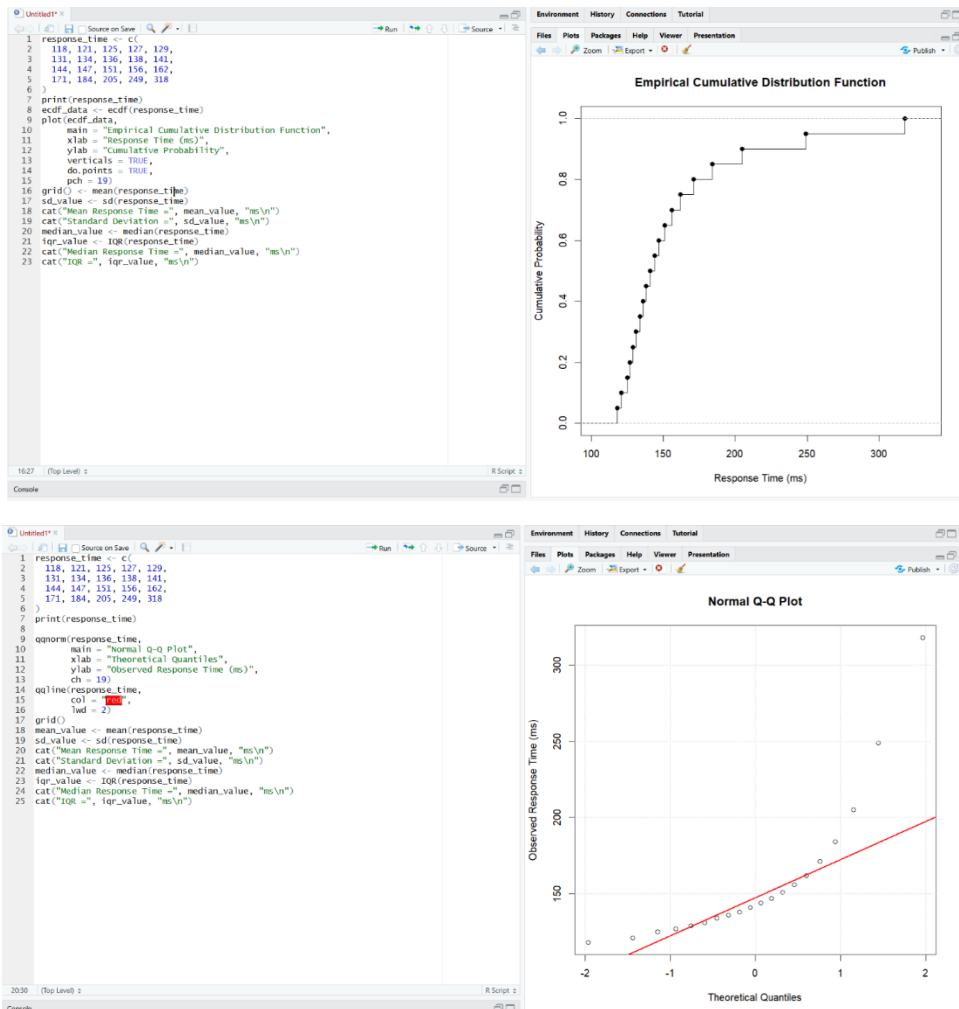
CO3_AT1_Assessment Tool: EXPERIMENT

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

Scenario: An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318

Tasks:



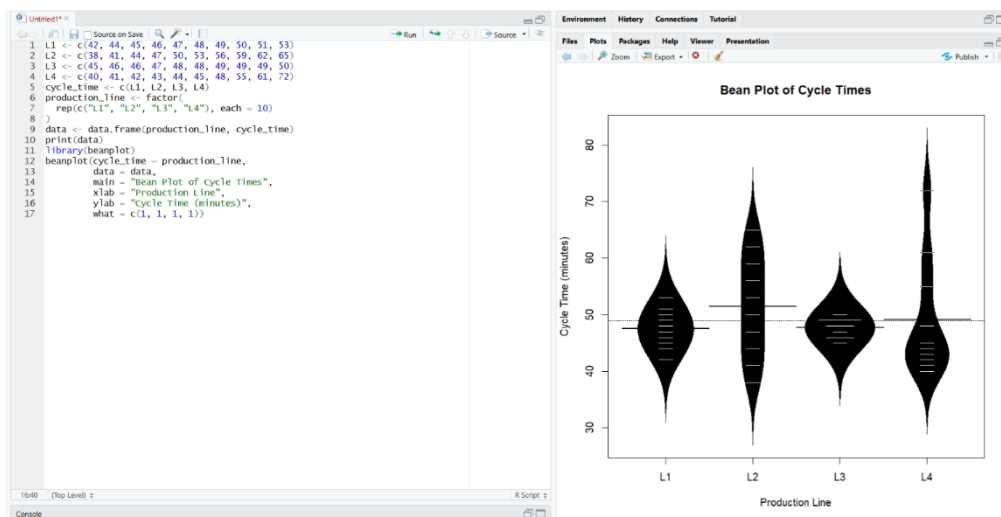
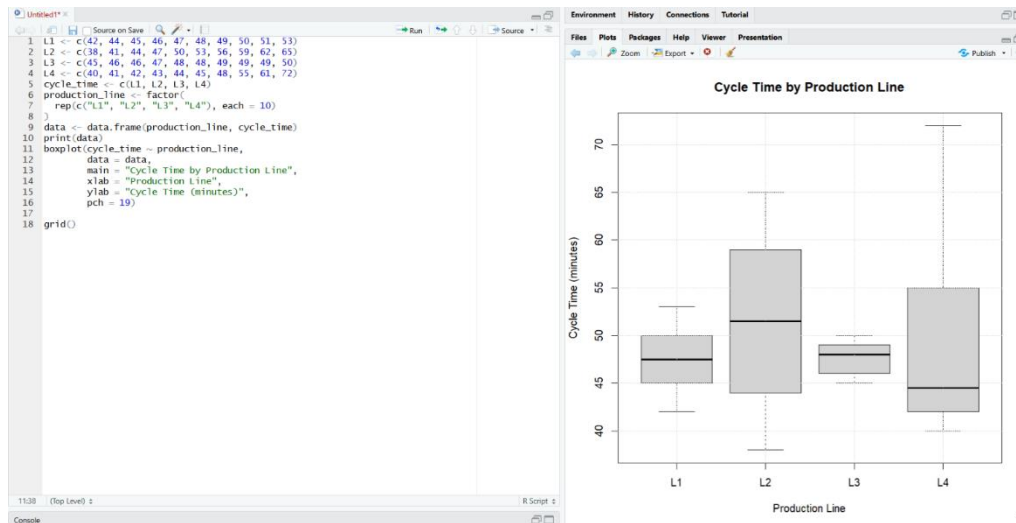
2. The ECDF and Q-Q plot show that the data are not perfectly normal, with noticeable deviation in the upper tail.
3. The data are right-skewed, mainly due to extreme values such as 249 ms and 318 ms.
4. Use percentile-based limits because the data are skewed and contain extreme observations; mean $\pm$ SD may be distorted.

Question 2 – Comparing Many Distributions and Bean Plots

Scenario: A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

Production Line	Cycle Times (minutes)
L1	42, 44, 45, 46, 47, 48, 49, 50, 51, 53
L2	38, 41, 44, 47, 50, 53, 56, 59, 62, 65
L3	45, 46, 46, 47, 47, 48, 48, 49, 49, 50
L4	40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks:

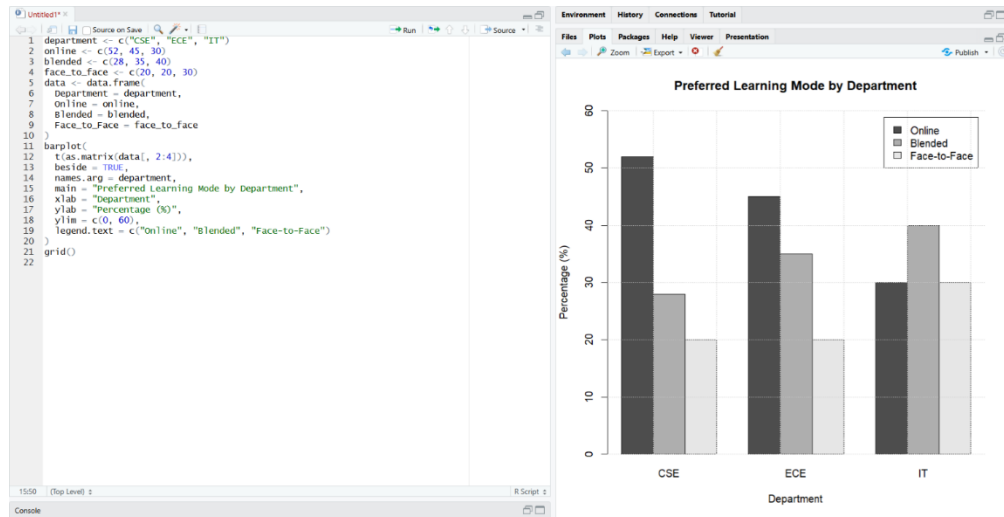


3. Greatest variability: L4; most concentrated distribution: L3.
4. Comparing only means is insufficient because lines can have similar means but very different spread, skewness, and extreme values.

Question 3 – Proportion Visualization and Choice of Chart

Scenario: A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20
ECE	45	35	20
IT	30	40	30



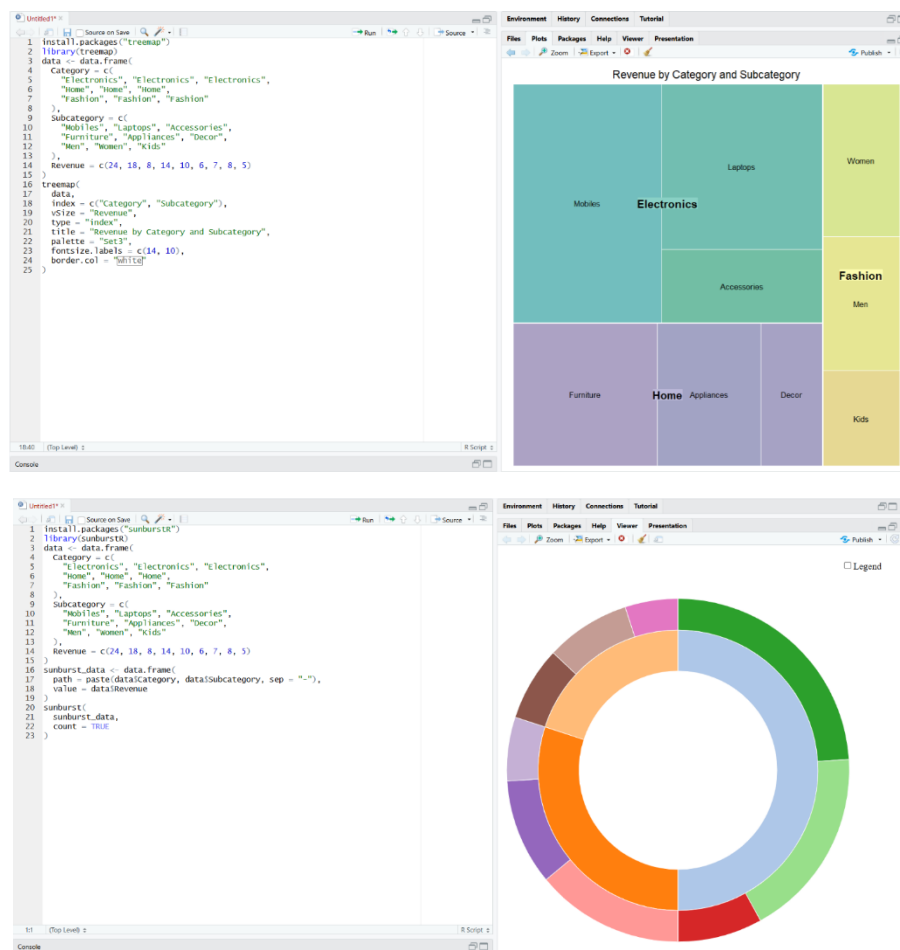
3. Side-by-side bar chart better supports comparison of individual proportions because all bars share a common baseline, making small differences easier to compare accurately.
4. Separate pie charts are less effective because comparing angles and areas across different pies is harder than comparing aligned bars.
5. A truncated y-axis (not starting at 0) can make small proportional differences appear much larger than they actually are.

Question 4 – Nested Proportions using Treemap and Sunburst

Scenario: An e-commerce company analyzes annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

Category	Subcategory	Revenue (₹ lakh)
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks:



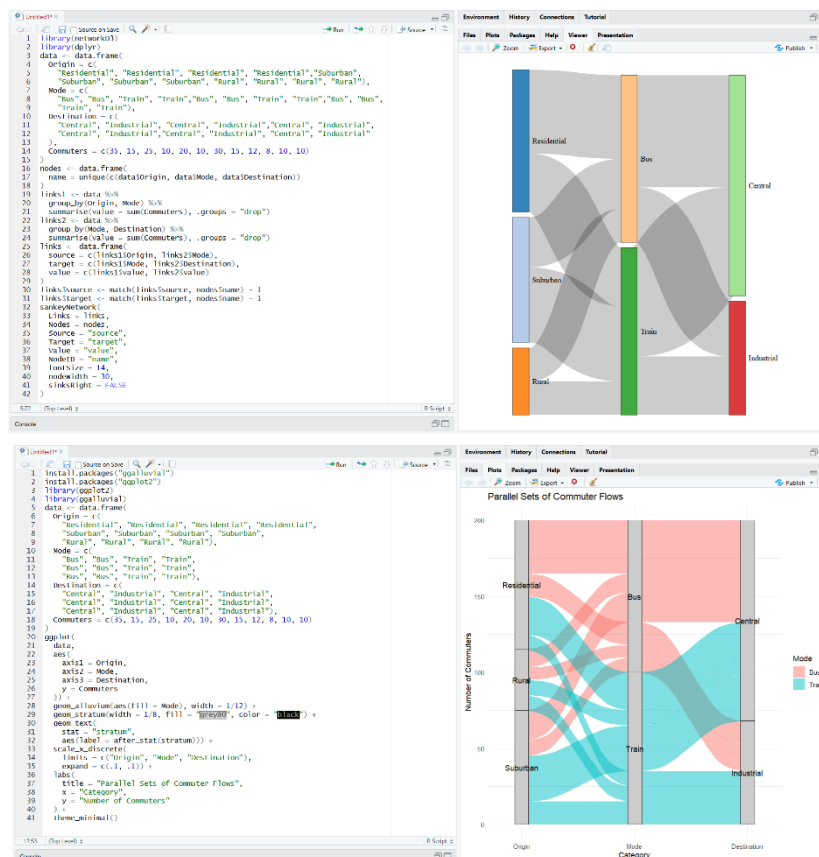
3. Largest category: Electronics with ₹50 lakh. Total revenue is ₹100 lakh, so Electronics contributes 50% of the total revenue.
4. Treemap vs Sunburst: A treemap is better for comparing sizes, while a sunburst is better for clearly showing the hierarchy between categories and subcategories.
5. Area encoding can mislead when two regions have similar areas, because humanKey competency: Treemaps, sunburst charts, nested proportions, area encoding

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets

Scenario: A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	Mode	Destination	Commuters
Residential	Bus	Central	35
Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Tasks:



- Three largest flows: Residential → Bus → Central = 35 (17.5%); Suburban → Train → Central = 30 (15%); Residential → Train → Central = 25 (12.5%).
- Sankey is better for showing flow magnitude, while parallel sets are better for comparing relationships between categorical groups.
- Poor link widths or category ordering can make a flow appear larger/smaller than it really is or make important relationships difficult to follow.

Suggested Evaluation Rubric – Experiment

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.